

MAIN PROJECT TITLE

PROJECT PRELIMINARY REPORT

**SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE AWARD OF DEGREE OF**

BACHELOR OF TECHNOLOGY

in

ELECTRICAL AND ELECTRONICS ENGINEERING

of

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

by

STD Name 1 (VAS21000)

STD Name 2 (VAS21000)

STD Name 3 (VAS21000)

STD Name 4 (VAS21000)



(AN ISO 9001:2015 CERTIFIED INSTITUTION)

Department of Electrical and Electronics Engineering

Vidya Academy of Science & Technology

Thalakkottukara, Thrissur - 680 501

(<http://www.vidyaacademy.ac.in>)

December 2026

Department of Electrical and Electronics Engineering Vidya Academy of Science & Technology

Thalakkottukara, Thrissur - 680 501

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(AN ISO 9001:2015 CERTIFIED INSTITUTION)

Certificate

This is to certify that the preliminary project report titled “**MAIN PROJECT TITLE**” is a bonafide record of the work carried out by **STUDENT NAME (VAS23EE000)** of Vidya Academy of Science & Technology, Thalakkottukara, Thrissur - 680 501 in partial fulfillment of the requirements for the award of **Degree of Bachelor of Technology** in **Electrical and Electronics Engineering** of **APJ Abdul Kalam Technological University**, during the academic year 2026-2027. The Preliminary Project report has been approved as it satisfies the academic requirements in the respect of preliminary project work prescribed for the said degree.

Project Guide

Guides Name

Asst. Prof., Dept. of EEE

Head of Department

Dr Vinila M. L.

Asso., Dept. of EEE

Project Co-guide

Co-guides Name

Asst. Prof., Dept. of EEE

Undertaking

I, **STUDENT NAME (Univ. Reg. No. VAS23EE000)**, hereby undertake that the preliminary project work entitled “**MAIN PROJECT TITLE**”, is carried out by me independently under the valuable guidance of **Guides Name**, Asst. Prof., and **Co-guides Name**, Asst. Prof., Dept. of Electrical and Electronics Engineering, Vidya Academy of Science and Technology, Thalakkottukara, Thrissur, in partial fulfillment of the requirements for the award of degree of **Bachelor of Technology in Electrical and Electronics Engineering** of **APJ Abdul Kalam Technological University**, during the academic year 2026-2027.

Thrissur

STUDENT NAME

December 2026



Institution Vision

To be a premier institution in engineering education and research, nurturing innovative professionals for sustainable societal development.

Institution Mission

M1: Provide quality engineering education that empowers students with strong technical competence and analytical problem solving abilities.

M2: Promote research, innovation, entrepreneurship and industry collaboration to develop sustainable technological solutions.

M3: Develop ethical and socially responsible professionals with leadership qualities, lifelong learning abilities and a commitment to societal well-being.

Department Vision

To be a reputed student centric department in electrical and electronics engineering, fostering innovative and ethically driven engineering professionals dedicated to sustainable societal development.

Department Mission

M1: Impart quality education in electrical and electronics engineering that equips students with strong theoretical knowledge, practical engineering skills, and problem-solving ability.

M2: Nurture a culture of research, innovation, and entrepreneurship in electrical and electronics Engineering for sustainable technological solutions.

M3: Develop professionally competent graduates with ethical integrity, leadership qualities, and social responsibility for lifelong learning and societal service.

Department PEOs

PEO-1: Graduates will apply strong foundational knowledge of mathematics, engineering sciences, and electrical and electronics engineering to formulate, analyze, and solve real-life problems in circuits, systems, and industrial installations.

PEO-2: Graduates will pursue research, innovation, and entrepreneurship, engaging in lifelong learning and higher studies to create sustainable and impactful technological advancements.

PEO-3: Graduates will demonstrate ethical integrity, leadership, effective communication, and multidisciplinary teamwork, contributing responsibly to societal and technological progress.

Department PSOs

PSO-1: Apply the fundamentals of electrical and electronics engineering, circuit theory, control, and instrumentation to analyze, design, and develop efficient electrical systems and solutions.

PSO-2: Integrate and apply knowledge of power systems, renewable energy, power electronics, electrical machines, drives, and electric vehicles to develop sustainable, economical, and industry-relevant solutions to electrical engineering problems.

PSO-3: Engage in continuous learning and apply modern engineering tools and advanced technologies to address emerging challenges in electrical and electronics engineering.

Department POs

Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/ develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multi-disciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Acknowledgement

During the course of my preliminary project work several persons collaborated directly and indirectly with me. Without their support it would be impossible for me to finish my work. That is why I wish to dedicate this section to recognize their support.

I want to start expressing my thanks to my project guide, **Guides Name**, Asst. Prof., Dept. of EEE & project Co-guide, **Co-guides Name**, Asst. Prof., Dept. of EEE, because of his valuable advice and guidance towards this work. I received motivation, encouragement and hold up from his during the course of work.

I would like to express my sincere gratitude to project coordinators **Dr. Neenu Thomas**, Asso. Prof. Dept. of EEE and **Mr Arun Xavier**, Asst. Prof.(Sr. Grade), of EEE department for their valuable suggestion and feedbacks.

I am grateful to express my thanks to all the faculty members of our department for their support. I articulate my gratitude to all my friends for their support and help for this work.

I am thankful to **Dr Vinila M. L.**, Head of Electrical and Electronics Engineering Department, and our Principal **Dr Sunitha C.**, for their sole co-operation.

Last, but not the least I wish to express my gratitude to God Almighty for His abundant blessings without which this effort would not have been successful.

STUDENT NAME

Reg. No. VAS23EE000

Seventh Semester B.Tech (2023 Admissions)

Vidya Academy of Science & Technology

December 2026

Thrissur - 680 501.

Abstract

Copy Paste Your Abstract here . . .

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List of Symbols and Abbreviations

ABCD	Use in alphabetical order
A_i	Area of the i^{th} component
DC	Direct current electricity
DSO	Digital Storage Oscilloscope
FBI	Full Bridge Inverter

Chapter 1

INTRODUCTION

Give some intro about this chapter

Electrical engineering is a field of engineering that generally deals with the study and application of electricity, electronics, and electromagnetism. This field first became an identifiable occupation in the latter half of the 19th century after commercialization of the electric telegraph, the telephone, and electric power distribution and use.[1] Photovoltaic (PV) power supplied to the system is gaining more and more visibility. [2]

Photovoltaic (PV) power supplied to the system is gaining more and more visibility Solid-state inverters have shown for the enabling technology for putting PV systems now. [2]

1.1 Objectives of the Work

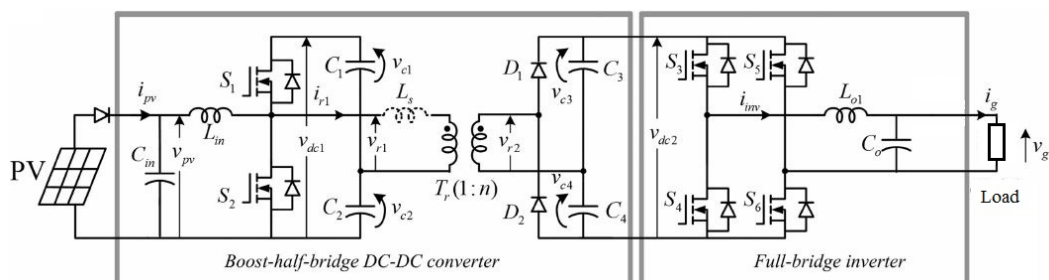


Figure 1.1: Circuit Diagram

The concept of micro-inverter (also known as a module integrated converter / inverter) has become a future trend for single phase photovoltaic (PV) power. In general, a micro-inverter PV is often provided by a low voltage solar panel, which requires a high voltage

lift ratio to the Fig: 1.1 produce the desired AC output .

1.2 Motivation for this work

There are . . .

1.3 Methodologies Adopted

Methodologies adopted for this project

1.4 Outline of the Report

This report contains 6 chapters. Chapter 1 gives the introduction to the project work and describes the objectives of the work. Literature survey is describes in Chapter 2. An introduction to the pro.....

1.5 Summary

Give summary abt this chapter & brief into about the next Chapter.

Chapter 2

LITERATURE REVIEW

Give some intro abt this chapter

2.1 Paper I [1]

Details about paper one. Here you write abt the details in the IEEE paper(Intro + Conclusion) or about the work who had done before, what are the problems at that time, etc.

2.2 Paper II[2]

Details about paper two. Here you write abt the details in the IEEE paper(Intro + Conclusion) or about the work who had done before, what are the problems at that time, etc

2.3 Paper III[3]

Details about paper three. Here you write abt the details in the IEEE paper(Intro + Conclusion) or about the work who had done before, what are the problems at that time, etc

2.4 Paper IV

Details about paper four. Here you write abt the details in the IEEE paper(Intro + Conclusion) or about the work who had done before, what are the problems at that time, etc

1. Paper I

2. Paper II

To use bullets

- Presented
- Paper
- Presented
- Paper

General, a micro-inverter PV is often provided by a low voltage solar panel, which requires a high voltage lift ratio to produce the desired AC output .

- Presented
- Paper
 - Presented
 - Paper
- Presented
- Paper

Inverter has become a future trend for single phase photovoltaic (PV) power. In general, a micro-inverter PV is often provided by a low voltage solar panel . . .

To use 1,2,3, numbers

1. prepare a source file with the extension "tex"
2. Compile it with $\text{\LaTeX}$ to produce a "dvi" file
3. Print the document using a "dvi" driver

For making a table . . .

Planet	Diameter(km)
Mercury	4878
Pluto	2274
Mercury	4878
Pluto	2274
Mercury	4878
Pluto	2274
Mercury	4878
Pluto	2274

Table 2.1: Table Name

$$ax + by + c = 0$$

2.5 Summary

Give summary abt this chapter & brief into about the next Chapter.

Chapter 3

METHODOLOGIES FOR THE PROJECT

Give some intro abt this chapter

3.1 Block Diagram

Explain different stages in the block diagram with figure . . .

3.2 Circuit Diagram

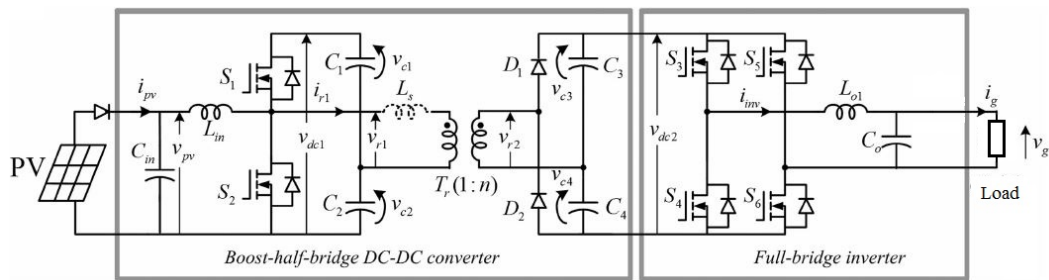


Figure 3.1: Circuit Diagram

Explain different stages in the circuit diagram with figure . . .

For equ..

$$d_f T_s = i_{pk} / m_2 = i_{pk} L_1 / (V_0 - v_{ik}) \quad (3.1)$$

During this switching cycle, the energy (E_k) transferred from the input to the output can be obtained as:

$$E_k = v_{ik} \cdot i_{pk} \cdot T_s (D + d_f) / 2 \quad (3.2)$$

3.3 Control Part

Explain different stages in the control part PIC, 8085, etc . . .

3.4 Power Circuit Part

The duty cycle of S1 is denoted by d_1 . The switching period of the pulse half-bridge converter is T_{sw1} . PV current and voltage are represented by I_{PV} and V_{PV} , respectively. The voltages across C1, C2, C3, and C4 are denoted by V_{C1} , V_{C2} , V_{C3} , V_{C4} , and, respectively. The transformer primary voltage, the secondary voltage and primary current are designated as v_{r1} , v_{r2} , and

Also explain basic parts in the project

Eg: FM Block, Relay, Voltage Doubler, etc . . . so explain each parts basic principle with appropriate figures . . .

3.5 Summary

Give summary abt this chapter & brief into about the next Chapter.

Chapter 4

HARDWARE SETUP OF PROJECT

Give some intro abt this chapter

4.1 Introduction

Hardware of the project with figures . . . and explanations λ

4.2 Hardware Setup of the Project

4.2.1 Power Circuit

4.2.2 Control Circuit

4.3 Hardware Components

4.3.1 First Compt

4.3.2 Second Compt

4.4 Complete Hardware Setup

Explain with figures

4.5 Discussions of the Experimental Results

4.6 Summary

Give summary abt this chapter & brief into about the next Chapter.

Chapter 5

Project status Completed in S7

A new

Chapter 6

Project status to be complected in S8

A new

Bibliography

- [1] Shuai Jiang, Dong Cao, Yuan Li, Member, and Fang Zheng Peng, "Boost-Half-Bridge Photovoltaic Micro inverter System Using Repetitive Current Control and Maximum Power Point Tracking," *IEEE Trans. Power Electron*, VOL. 27, no. 11, pp. 4711-4722, Nov. 2012.
- [2] C. Yoon, J. Kim, and S. Choi, "A Review of Single-Phase Grid-Connected Inverters for Photovoltaic Modules," *IEEE Trans. Industry Applications*, vol. 41, no. 5, pp. 1292–1306, Set/Oct. 2005.
- [3] S. B. Kjaer, J. K. Pedersen, and F. Blaabjerg, "Multiphase DC – DC converters using a boost-half-bridge cell for high-voltage and high-power applications," *IEEE Trans. Power Electron.*, vol. 26, no. 2, pp. 381–388, Feb. 2011.

APPENDIX

Photovoltaic (PV) power supplied to the system is gaining more and more visibility while the demand for power in compared to traditional energy sources like gas, oil, coal, hydro wind, etc. . . .

Pgm Code for the IC's

Datasheet of IC's

Datasheet of MOSFETs

CODE FOR THE BOOST-HALF-BRIDGE CONVERTER PULSES

```
LCD_BUS      REG  P0      ;LCD DATA BUS
H_DRIVE      REG  P1.0    ;
L_DRIVE      REG  P1.1    ;
COUNT       EQU  12H

;-----

ORG  000H
SJMP RESET           ; POWER UP RESET VECTOR
ORG  003H
AJMP EXINT0          ; EXTERNAL INTERRUPT 0 VECTOR
ORG  00BH
AJMP TIMER0          ; COUNTER/ TIMER 0 INT VECTOR
ORG  013H
AJMP EXINT1          ; EXTERNAL INT 1 VECTOR
ORG  01BH        ;
AJMP TIMER1          ; TIMER 1 INT VECTOR
ORG  023H
AJMP SERIAL          ; SERIAL INT VECTOR
ORG  0050H

RESET:
MOV  SP,#5FH         ;STACK POINTER VECTRED TO 5FH
CLR  H_DRIVE         ;
SETB L_DRIVE

MAIN:
ACALL DRIVE_CONTROL
      SJMP MAIN
```

;-----DRIVE CONTROL-----

DRIVE_CONTROL:

MOV COUNT,#4 ;OFF TIME

NOP

NOP

LOOP_OFF:

CLR H_DRIVE

NOP

SETB L_DRIVE

NOP

DJNZ COUNT,LOOP_OFF

MOV COUNT,#10 ;ON TIME

LOOP_ON:

CLR L_DRIVE

NOP

SETB H_DRIVE

NOP

DJNZ COUNT,LOOP_ON

RET

;-----

TIMER0:

EXINT1:

EXINT0:

TIMER1:

SERIAL:

RETI



IRF830

N - CHANNEL 500V - 1.35Ω - 4.5A - TO-220 PowerMESH™ MOSFET

TYPE	V _{DSS}	R _{DS(on)}	I _D
IRF830	500 V	< 1.5 Ω	4.5 A

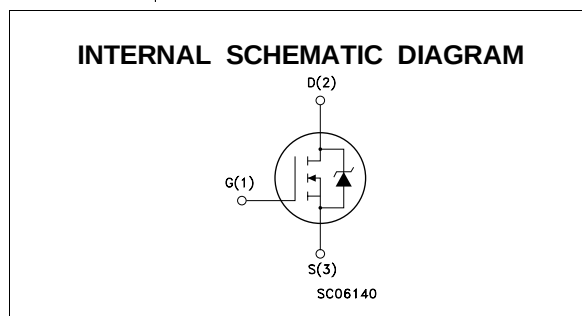
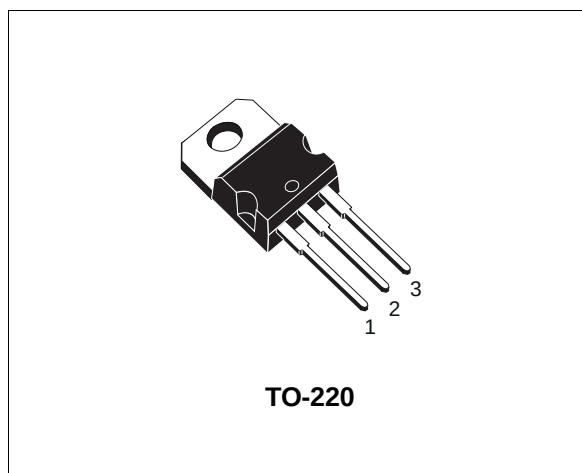
- TYPICAL R_{DS(on)} = 1.35 Ω
- EXTREMELY HIGH dv/dt CAPABILITY
- 100% AVALANCHE TESTED
- VERY LOW INTRINSIC CAPACITANCES
- GATE CHARGE MINIMIZED

DESCRIPTION

This power MOSFET is designed using the company's consolidated strip layout-based MESH OVERLAY™ process. This technology matches and improves the performances compared with standard parts from various sources.

APPLICATIONS

- HIGH CURRENT, HIGH SPEED SWITCHING
- SWITCH MODE POWER SUPPLIES (SMPS)
- DC-AC CONVERTERS FOR WELDING EQUIPMENT AND UNINTERRUPTIBLE POWER SUPPLIES AND MOTOR DRIVER



ABSOLUTE MAXIMUM RATINGS

Symbol	Parameter	Value	Unit
V _{DS}	Drain-source Voltage (V _{GS} = 0)	500	V
V _{DGR}	Drain- gate Voltage (R _{GS} = 20 kΩ)	500	V
V _{GS}	Gate-source Voltage	± 20	V
I _D	Drain Current (continuous) at T _c = 25 °C	4.5	A
I _D	Drain Current (continuous) at T _c = 100 °C	2.9	A
I _{DM} (•)	Drain Current (pulsed)	18	A
P _{tot}	Total Dissipation at T _c = 25 °C	100	W
	Derating Factor	0.8	W/°C
dv/dt(1)	Peak Diode Recovery voltage slope	3.5	V/ns
T _{stg}	Storage Temperature	-65 to 150	°C
T _j	Max. Operating Junction Temperature	150	°C

(•) Pulse width limited by safe operating area

(1) I_{SD} ≤ 4.5A, di/dt ≤ 75 A/μS, V_{DD} ≤ V_{(BR)DSS}, T_j ≤ T_{JMAX}

First Digit of the Datecode Being Z or K Identifies Silicon Characterized in this Datasheet

THERMAL DATA

R _{thj-case}	Thermal Resistance Junction-case	Max	1.25	°C/W
R _{thj-amb}	Thermal Resistance Junction-ambient	Max	62.5	°C/W
R _{thc-sink}	Thermal Resistance Case-sink	Typ	0.5	°C/W
T _l	Maximum Lead Temperature For Soldering Purpose		300	°C

AVALANCHE CHARACTERISTICS

Symbol	Parameter	Max Value	Unit
I _{AR}	Avalanche Current, Repetitive or Not-Repetitive (pulse width limited by T _j max)	4.5	A
E _{AS}	Single Pulse Avalanche Energy (starting T _j = 25 °C, I _D = I _{AR} , V _{DD} = 50 V)	290	mJ

ELECTRICAL CHARACTERISTICS (T_{case} = 25 °C unless otherwise specified)

OFF

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
V _{(BR)DSS}	Drain-source Breakdown Voltage	I _D = 250 µA V _{GS} = 0	500			V
I _{DSS}	Zero Gate Voltage Drain Current (V _{GS} = 0)	V _{DS} = Max Rating V _{DS} = Max Rating T _c = 125 °C			1 50	µA µA
I _{GSS}	Gate-body Leakage Current (V _{DS} = 0)	V _{GS} = ± 20 V			± 100	nA

ON (*)

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
V _{GS(th)}	Gate Threshold Voltage	V _{DS} = V _{GS} I _D = 250 µA	2	3	4	V
R _{DS(on)}	Static Drain-source On Resistance	V _{GS} = 10V I _D = 2.7 A		1.35	1.5	Ω
I _{D(on)}	On State Drain Current	V _{DS} > I _{D(on)} × R _{DS(on)max} V _{GS} = 10 V	4.5			A

DYNAMIC

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
g _{fs} (*)	Forward Transconductance	V _{DS} > I _{D(on)} × R _{DS(on)max} I _D = 2.7 A	2.5			S
C _{iss} C _{oss} C _{rss}	Input Capacitance Output Capacitance Reverse Transfer Capacitance	V _{DS} = 25 V f = 1 MHz V _{GS} = 0		610 120 10		pF pF pF

ELECTRICAL CHARACTERISTICS (continued)

SWITCHING ON

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
$t_{d(on)}$	Turn-on Time	$V_{DD} = 250\text{ V}$ $I_D = 2.9\text{ A}$		11.5		ns
t_r	Rise Time	$R_G = 4.7\ \Omega$ $V_{GS} = 10\text{ V}$ (see test circuit, figure 3)		8		ns
Q_g	Total Gate Charge	$V_{DD} = 400\text{ V}$ $I_D = 3\text{ A}$ $V_{GS} = 10\text{ V}$		22	30	nC
Q_{gs}	Gate-Source Charge			7.2		nC
Q_{gd}	Gate-Drain Charge			8		nC

SWITCHING OFF

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
$t_{r(Voff)}$	Off-voltage Rise Time	$V_{DD} = 400\text{ V}$ $I_D = 4.5\text{ A}$		7		ns
t_f	Fall Time	$R_G = 4.7\ \Omega$ $V_{GS} = 10\text{ V}$ (see test circuit, figure 5)		5		ns
t_c	Cross-over Time			15		ns

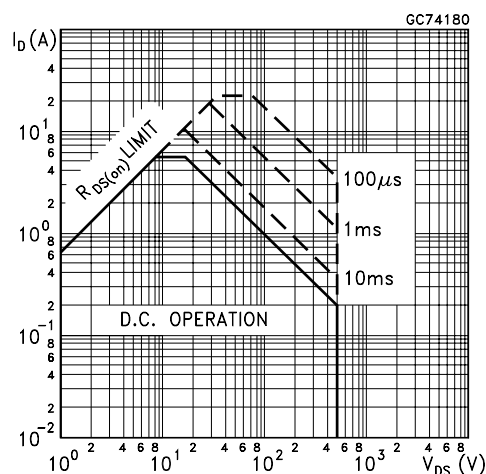
SOURCE DRAIN DIODE

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
I_{SD}	Source-drain Current				4.5	A
$I_{SDM}(\bullet)$	Source-drain Current (pulsed)				18	A
$V_{SD} (*)$	Forward On Voltage	$I_{SD} = 4.5\text{ A}$ $V_{GS} = 0$			1.6	V
t_{rr}	Reverse Recovery Time	$I_{SD} = 4.5\text{ A}$ $di/dt = 100\text{ A}/\mu\text{s}$ $V_{DD} = 100\text{ V}$ $T_j = 150\text{ }^\circ\text{C}$ (see test circuit, figure 5)		435		ns
Q_{rr}	Reverse Recovery Charge			3.3		μC
I_{RRM}	Reverse Recovery Current			15		A

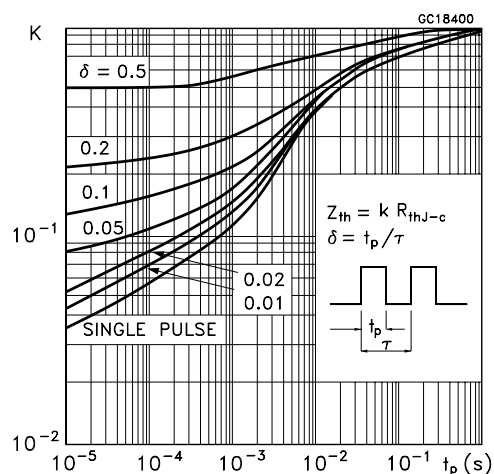
(*) Pulsed: Pulse duration = 300 μs , duty cycle 1.5 %

($\bullet$) Pulse width limited by safe operating area

Safe Operating Area



Thermal Impedance





Department of Electrical and Electronics Engineering
Vidya Academy of Science & Technology
Thalakkottukara, Thrissur - 680 501
(<http://www.vidyaacademy.ac.in>)